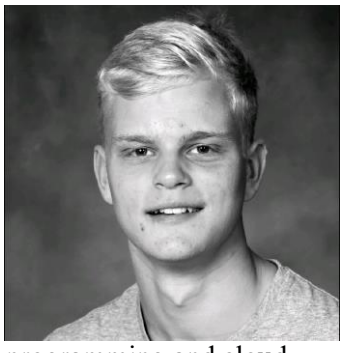
Erik Frankling



Esplananden 30
183 39 Täby
Tel: +46 76-838 38 91
E-post: erik.frankling@frankling.se
linkedIn Github

I am a Computer Science student at KTH with a strong interest in systems programming and cloud-native technologies. I have continuously engaged in various personal programming projects since my early teenage years, and recently gained hands-on experience with cutting-edge technologies during my internship at Ericsson's research department.

Work experience

Software Engineering Intern, Ericsson June 2025 – October 2025

Designed and implemented a real-time compute offloading application for 5G networks, processing real-time 5G performance metrics to make intelligent compute offloading decisions about when and to where to offload using Kubernetes, InfluxDB, Golang, and Python

Deployed and configured 5G network monitoring infrastructure at Lund Technical University's robotics laboratory as part of AORTA, a national collaborative research initiative, leveraging prior expertise with the data collection pipeline to integrate the system into their experimental 5G environment

Ported Rust and C applications to Go, demonstrating polyglot interoperability in WASI components and contributing to TinyGo compiler development through bug identification and resolution

Sailing instructor, Vitsgarn Sailing School Summer 2021 – 2024

Leader of sailing camps for children and teenagers.

Education

Master of Science in Computer Science and Engineering August 2023 – ongoing
KTH Royal Institute of Technology – KTH
Courses in mathematics, programming, writing, product design and project organization. Participated in the most advanced programming group (the "++" group) comprising the top 10% of the year's students.

Natural Sciences Programme August 2020 – June 2023
Östra Reals Gymnasium
High school education in the natural sciences program, including several extended university-level mathematics courses.
Final grade average: 21.65

Engelbrektsskolan year 7-9, Advanced education in mathematics August 2017 – June 2020
Specialized national programme in advanced mathematics.

University Entrance Exam – HP (Spring 2023): 1.8

Projects

Cloud Comfort – Agentic Cloud Infrastructure Dev-Platform 2026

2026 HackEurope hackathon, team of 3

Built a development platform for cloud deployments powered by agentic AI. Users describe their desired infrastructure in natural language, and the tool autonomously generates Terraform files, validates them, and deploys to AWS/GCP/Azure. Supports multiple LLM providers including Claude and OpenRouter. Includes interactive infrastructure visualization through Terraform-to-diagram pipeline. Go backend, React/TypeScript frontend.

Voice Streaming Application – Purrcast

Part of the KTH course: Software Engineering in Project Form 2025

Developed a web-based intercom system based on request from a "client" at KTH to enable real-time backstage communication during live performances. Served as lead backend developer in an 8-person team, implementing Go backend with WebRTC audio streaming and WebSocket signaling. Designed SFU (Selective Forwarding Unit) topology for multi-channel audio routing and coordinated API design with React frontend team.

Raytracing on GPU and CPU 2023 – 2024

Part of KTH First year programming course

Created two alternative raytraced rendering implementations - on GPU with the WebGPU API as well as a CPU-based algorithm.

Declarative Linux Environment – NixOS Configuration 2023 – ongoing

Personal project

Maintain a comprehensive NixOS configuration managing system packages, development environments, and dotfiles through declarative configuration. Includes custom Neovim setup with LSP integrations, window manager configuration, and reproducible development shells. Demonstrates infrastructure-as-code practices and deep Linux system administration knowledge.

Ergonomic keyboard made with 3D printer 2022 – ongoing

Personal project

Designed a more ergonomic keyboard optimized for programming. Printed on my 3D printer and handsoldered, used daily. Dactyl Manuform with Gazzew U4 Silent Tactile Switches, Arduino Pro Micro

Evolutionary AI simulation 2022 – 2023

High school final project

Competed in the national Young Researchers ("Unga Forskare") competition, reaching semifinals. Implemented an algorithm for evolutionary AI learning in an ecosystem simulation where creatures develop better survival strategies. Written in C# with Unity for graphics.

Languages: English, Swedish

Languages & Technologies: Rust, C, C++, Go, Python, JavaScript, C#, Java, Haskell, Prolog, Zig, SQL, Bash, WebSockets, WebAssembly (WASM/WASI), Kubernetes, Docker, Helm, Linux, Git, InfluxDB, PostgreSQL, Grafana, MQTT, Embedded Systems, Unity, Vite, WebGPU, WebRTC, 3D-printing, Agentic AI Agentic Engineering, ML, LLM, Claude Code, OpenCode, Codex, Terraform, AWS, Cloud Native, GitHub, CI/CD